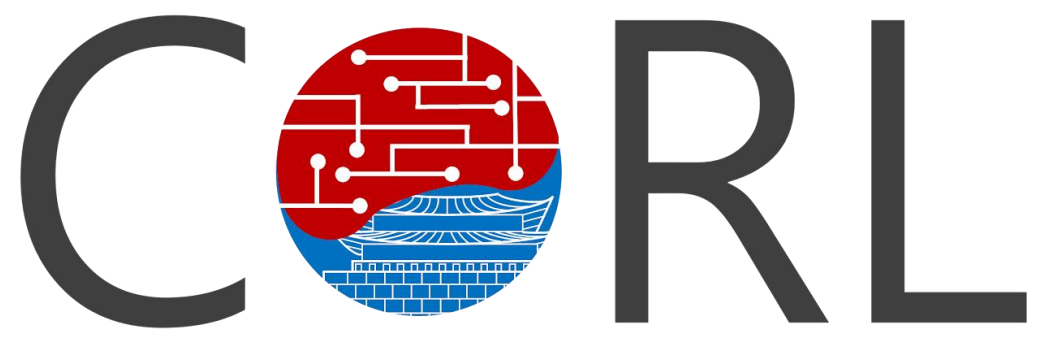


Contrastive In-Context Learning with Active Memory for Task Planning



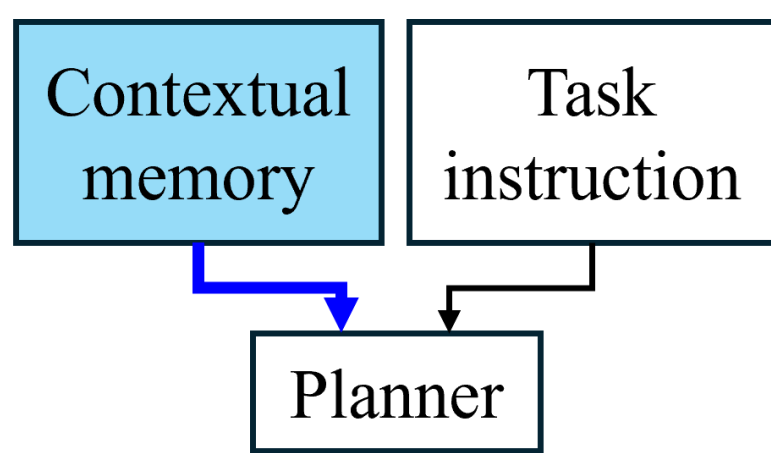
Jiho Lee¹ Kyungjae Lee² Eunwoo Kim¹
Chung-Ang University¹ Korea University²



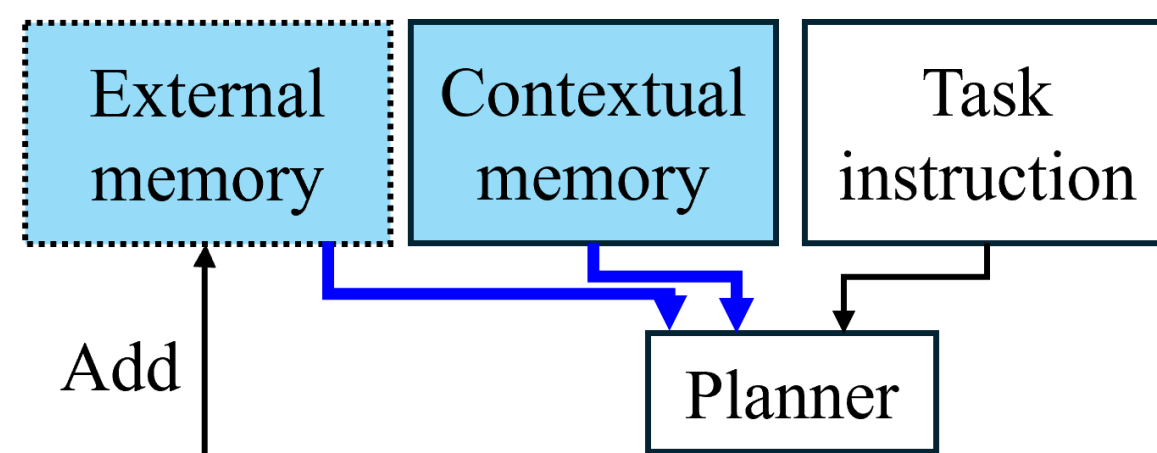
Introduction & Motivation

Standard in-context learning

- The model (LLM) generates task plans using a fixed contextual memory without additional fine-tuning
- Contextual memory
 - Composed of (a) a few input-output successful demonstrations manually curated by human experts for positive prompting



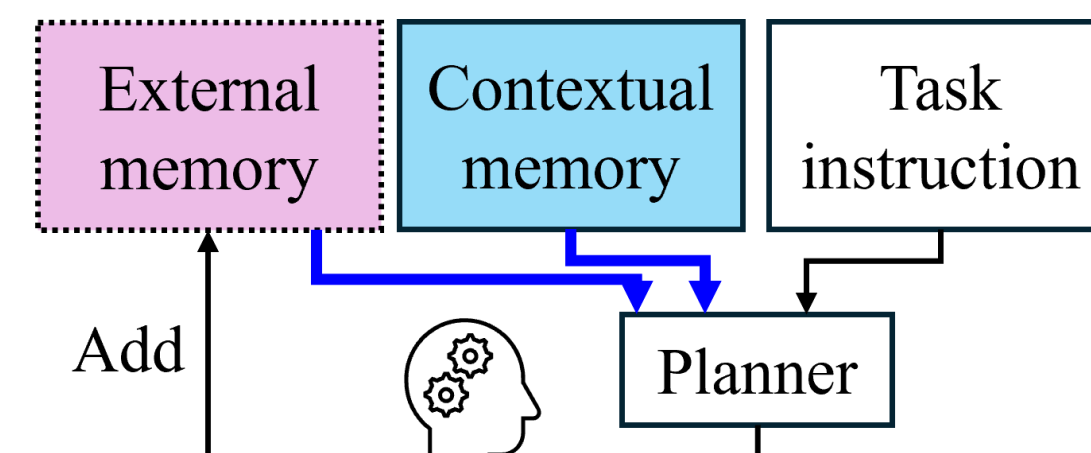
(a) Contextual memory



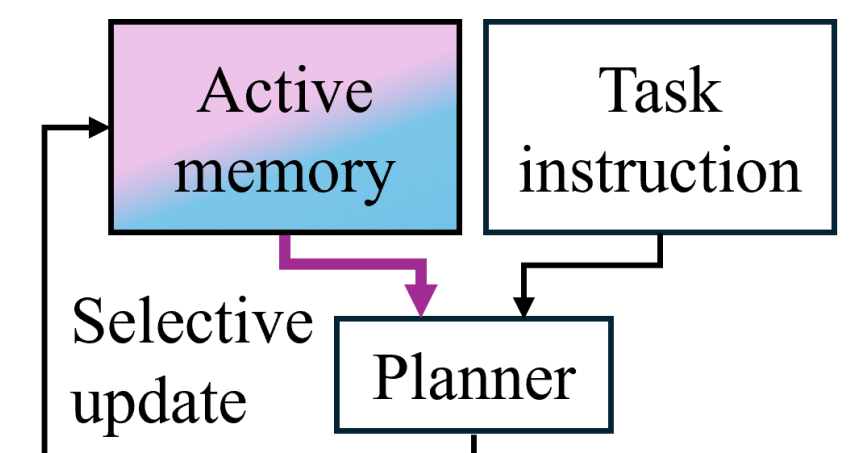
(b) External success memory

In-context learning with external memory

- The model (LLM) generates task plans using both contextual memory and external memory, then the generated plans are stored in external memory
- External memory
 - Composed of either (b) successful demonstrations or (c) corrected failures for positive prompting



(c) External failure memory



(d) Ours

Research challenges

- 1) Learn with positive prompting to imitate the provided examples in memory $\Rightarrow$ **Contrastive in-context learning**
- 2) Use static or ever-expanding memory without memory management $\Rightarrow$ **Active memory**

Proposed Method

Contrastive in-context learning via dual prompting

- Learn by exploiting the contrast between positive prompting from successes to guide desirable reasoning patterns and negative prompting from failures to help in recognizing incorrect behaviors
- Design
 - Instruction signal g + demonstration $d_i = (x_i, y_i)$
- Example of dual prompting

Positive prompting (g_p, x_i, y_i)

This is a *success and correct* plan. *Learn* this example and *follow* the pattern and logic demonstrated.

- Initial state: {(empty cup1)} • Goal state: {(fill cup1 coffee)}
- Plan: {(pick cup1 table) (place cup1 coffee-machine) (*toggle-on coffee-machine*) (*toggle-off coffee-machine*) (pick cup1 coffee-machine) (place cup1 table)}

Negative prompting (g_n, x_i, y_i)

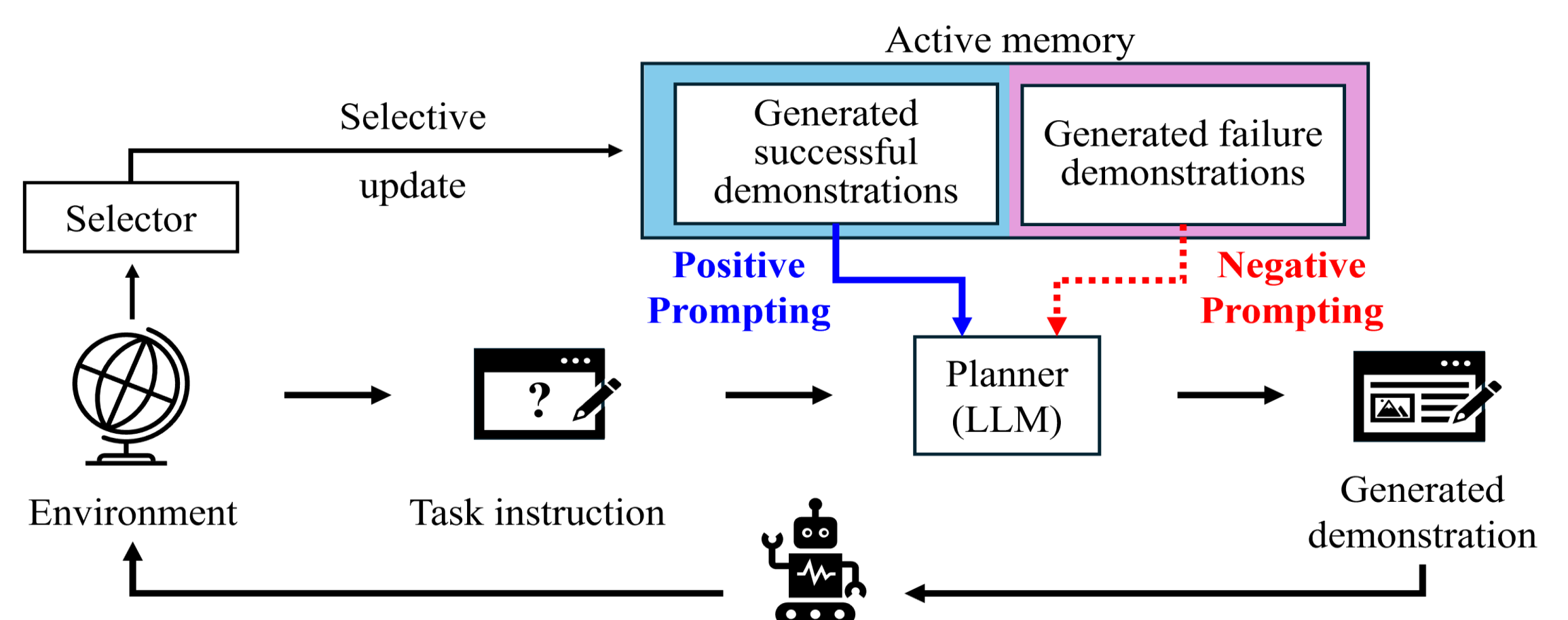
This is a *failure and incorrect* plan. *Do not learn* this example and *avoid* the pattern and logic shown.

- Initial state: {(empty cup1)} • Goal state: {(fill cup1 coffee)}
- Plan: {(pick cup1 table) (place cup1 coffee-machine) (*toggle-off coffee-machine*) (pick cup1 coffee-machine) (place cup1 table)}

Active memory

- Serve as self-evolving repository of LLM-generated demonstrations by selectively integrating useful ones
- Diversity-aware selection strategy
 - Leverage language embedding with Sentence-BERT and k-means clustering to capture semantic and contextual features of demonstrations

Proposed framework



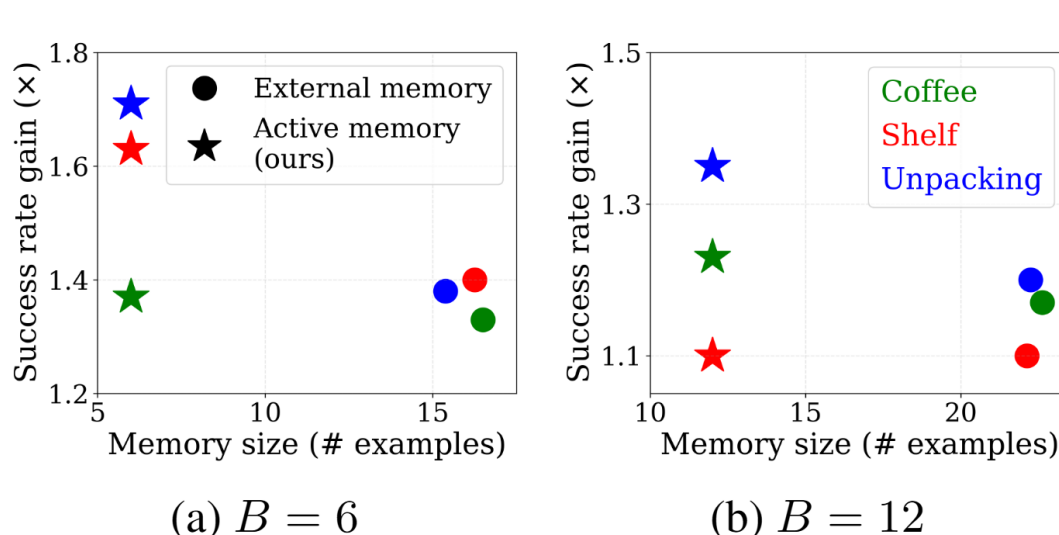
Experimental Results

Benchmark experiments

- Comparison of task success rates

Method	Shelf		Coffee		Unpacking		Avg.
	$N = 5$	$N = 7$	$N = 2$	$N = 4$	$N = 5$	$N = 7$	
GPT-4o-mini							
Standard planning	72.5%	35.0%	77.5%	62.5%	62.5%	32.5%	57.1%
Re-planning	90.0%	37.5%	85.0%	77.5%	67.5%	37.5%	65.8%
Planning w/ external success memory	95.0%	50.0%	90.0%	85.0%	77.5%	45.0%	73.8%
Planning w/ external failure memory	90.0%	47.5%	87.5%	82.5%	77.5%	52.5%	72.9%
Planning w/ active memory (ours)	97.5%	52.5%	90.0%	87.5%	82.5%	57.5%	77.9%
Gemini-1.5-Flash-8B							
Standard planning	52.5%	20.0%	70.0%	57.5%	35.0%	25.0%	43.3%
Re-planning	55.0%	20.0%	85.0%	55.0%	37.5%	22.5%	45.8%
Planning w/ external success memory	57.5%	30.0%	90.0%	80.0%	47.5%	25.0%	55.0%
Planning w/ external failure memory	65.0%	22.5%	85.0%	67.5%	52.5%	20.0%	52.1%
Planning w/ active memory (ours)	67.5%	27.5%	90.0%	80.0%	60.0%	35.0%	60.0%

Efficiency of active memory



Prompt ratio

Prompt ratio	0%	33%	50%	67%
B = 6				
Easy	-	+4.2%	+11.7%	+5.8%
Hard	-	+5.0%	+8.3%	+1.7%
B = 12				
Easy	-	+0.8%	+4.2%	+1.7%
Hard	-	+2.5%	+3.3%	+1.7%

Ablation study

Functionality			Shelf		Coffee		Unpacking		Avg.
Active memory	Dual prompting	Selector	$N = 5$	$N = 7$	$N = 2$	$N = 4$	$N = 5$	$N = 7$	
GPT-4o-mini									
			72.5%	35.0%	77.5%	62.5%	62.5%	32.5%	57.1%
✓			82.5%	40.0%	85.0%	75.0%	72.5%	45.0%	66.7%
✓		✓	92.5%	47.5%	87.5%	80.0%	72.5%	50.0%	71.7%
✓	✓		90.0%	45.0%	87.5%	82.5%	75.0%	50.0%	71.7%
✓	✓	✓	97.5%	52.5%	90.0%	87.5%	82.5%	57.5%	77.9%
Gemini-1.5-Flash-8B									
			52.5%	20.0%	70.0%	57.5%	35.0%	25.0%	43.3%
✓			60.0%	22.5%	80.0%	72.5%	45.0%	27.5%	51.3%
✓		✓	62.5%	25.0%	80.0%	75.0%	47.5%	27.5%	52.9%
✓	✓		65.0%	25.0%	85.0%	75.0%	52.5%	32.5%	55.8%
✓	✓	✓	67.5%	27.5%	90.0%	80.0%	60.0%	35.0%	60.0%

Real-world experiments



Please send any comments or questions to j2hoooo@cau.ac.kr